Shahriar Hossain

Software Engineer · Machine Learning & Full-Stack Development · Competitive Programmer

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SUMMARY

Computer Science student at CUET with hands-on professional experience as a Full-Stack Developer and a growing focus on Machine Learning. Strong competitive programming background (500+ problems across Codeforces, LeetCode, and others) with solid knowledge of algorithms, data structures, OOP, and MVC patterns. Experienced in Python, C++, and JavaScript. Published NLP researcher at ACL 2025 (7th place, CQs-Gen shared task) and IEEE ICCIT 2025. Eager to contribute to an innovative ML team and grow within a product-driven, collaborative environment.

PROFESSIONAL EXPERIENCE

Full-Stack Engineer Jan 2026 – Present

Mudraa · Dhaka, Remote

- Designed and implemented end-to-end web features for a financial product comparison platform, building scalable frontend interfaces and backend APIs using React, Node.js, and Firestore.
- Integrated dynamic investment data pipelines and collaborated with product and design teams to deliver production-ready full-stack solutions; applied Python-based scripting for data processing tasks.

Software Engineer Intern May – Dec 2025

Poridhi.io · Remote

- Designed scalable RESTful APIs with Node.js and implemented secure OAuth2 / OpenID Connect authentication; worked with DevOps team on Docker and GitHub Actions CI/CD pipelines.
- Applied ML-adjacent concepts including API integration for AI services; collaborated asynchronously across distributed teams with structured written documentation.

Frontend Developer Nov 2024 – Mar 2025

The Tork · On-Site

- Developed modular, responsive UI components with React.js, Vue.js, and Next.js for a high-traffic marketplace; sprint and code review management via Jira and Bitbucket.

EDUCATION

B.Eng. Computer Science & Engineering Mar 2022 – Present

Chittagong University of Engineering & Technology (CUET)

Relevant Coursework: Software Engineering, Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Information Security, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, Java, SQL

ML & AI: Machine Learning fundamentals, NLP (LSTM, DistilBERT, LLMs), Computer Vision concepts, RAG systems, LLM API integration (Anthropic, OpenAI, OpenRouter, Gemini), OCR, TTS

Algorithms & DS: 500+ competitive programming problems — Codeforces, LeetCode, UVA, LightOJ, CodeChef; strong in graphs, DP, trees, sorting

OOP & Frameworks: OOP principles, MVC pattern, React.js, Next.js, Vue.js, Node.js, Express, FastAPI

DevOps & Tools: Git, Docker, GitHub Actions, CI/CD, Jira, Bitbucket, ESLint, Firebase, AWS (S3, Lambda, SageMaker)

Databases: Firestore, PostgreSQL, REST API design, OAuth2, OIDC, JWT

PROJECTS

nano-vLLM Architectural Lab [GitHub] Feb 2026

- Step-by-step architectural deep-dive into vLLM internals — GPU underutilization (60–70% idle), memory fragmentation (30–50% waste), and throughput bottlenecks with enhanced visualizations.
- Hands-on exploration of LLM inference optimization techniques; directly relevant to ML engineering and production model serving.

Recommender System [GitHub] Dec 2025

- Python-based recommendation engine implementing collaborative filtering and content-based techniques; demonstrates applied ML pipeline design, data processing, and model evaluation.

eKYC Platform [GitHub] Nov 2025

- Electronic Know Your Customer (eKYC) platform with identity verification workflows — integrates computer vision concepts for document and face verification; full-stack JavaScript with REST API backend.

FastAPI Connection Pool MVP [GitHub] Dec 2025

- High-performance Python FastAPI backend with optimized PostgreSQL connection pooling — 100% success rate under 100+ concurrent requests; clean architecture, OOP design, MIT licensed.

Kong + Authentik OIDC API Gateway [GitHub] May 2025

- End-to-end API security setup with Kong API Gateway and Authentik as OIDC identity provider — automated authentication, access control, and route protection for microservices.

RESEARCH & PUBLICATIONS

Mind_Matrix at CQs-Gen 2025: Adaptive Generation of Critical Questions for Argumentative Interventions Jul 2025

12th Argument Mining Workshop (ArgMining 2025), ACL Anthology

Built an LLM pipeline (quantized model + semantic similarity + meta-evaluation loop) to generate critical questions challenging logical fallacies and implicit assumptions in argumentative text. Ranked 7th internationally.

Mental Health Detection from Journaling Data Using LSTM and DistilBERT-Based Lightweight NLP Models Dec 2025

IEEE ICCIT 2025 — International Conference on Computer and Information Technology

Proposed a hybrid NLP framework (LSTM + DistilBERT) for mental health risk detection from journal-style text. Integrated SHAP and LIME explainability tools to build transparent, trustworthy AI for healthcare.

CERTIFICATIONS & TRAINING

Front-End Development (React / Node.js / Vue.js / Angular.js) Apr 2025

Bangladesh Computer Council

Industrial Attachment — SELISE Bangladesh Ltd. (Ref: SELISEBD/CUET/25733) Nov 2025

Mentored by senior software development professionals, Dhaka

AWARDS & ACHIEVEMENTS

7th Place — ACL 2025 CQs-Gen Shared Task · ArgMining Workshop, Vienna, Austria — Team Mind_Matrix, first-ever shared task participation.

Ranked 21st — NAACL 2025 NLP Shared Task · Top 25 worldwide in competitive NLP research.

Competitive Programming · 500+ problems solved — Codeforces, LeetCode, UVA, LightOJ, CodeChef — strong algorithmic problem-solving background.

FIELD OF INTEREST

Machine Learning, Artificial Intelligence, Computer Vision, Natural Language Processing, LLM Inferencing, RAG Systems, Watermarking for LLMs, OCR Systems, TTS Systems, Algorithmic Problem Solving.